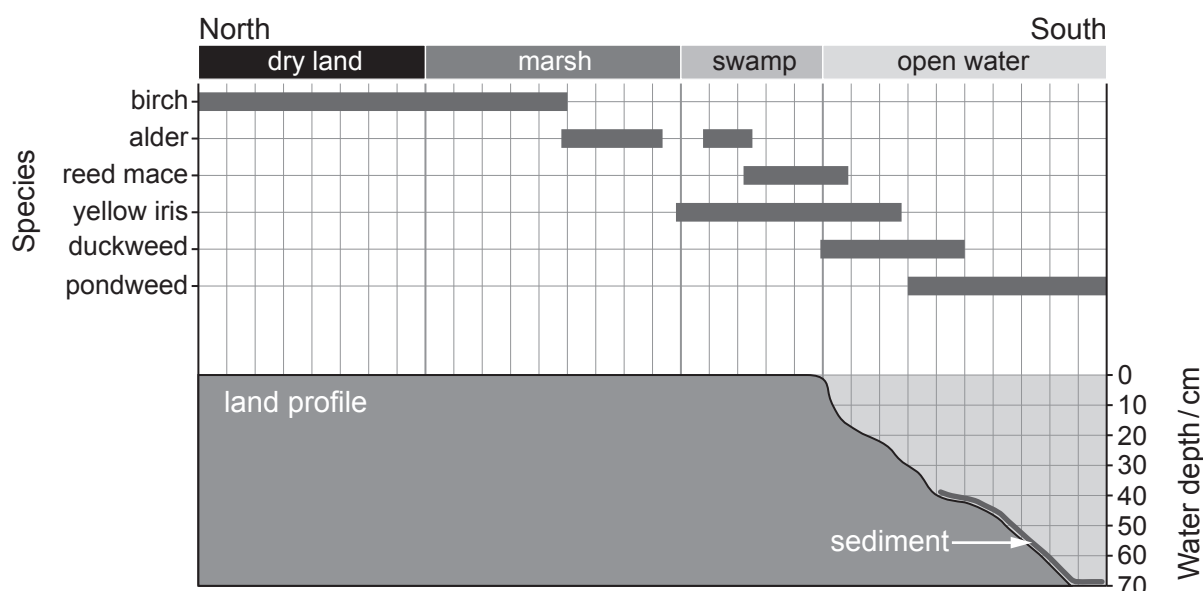


6. Wetland ecosystems are increasingly threatened as land is drained for agriculture. **Image 6.1** shows some of the results of a survey carried out at Valley Wetlands Centre, Anglesey.

Image 6.1



In order to carry out this survey, the team laid a rope across the area from north to south and identified the plant species that touched the rope.

- (a) (i) State the name given to the type of sampling technique used in this survey. [1]

.....

- (ii) The swamp in this survey could be correctly termed as both a habitat and an ecosystem. Distinguish between the terms, habitat and ecosystem. [2]

.....

.....

.....

- (iii) Using the information provided in **Image 6.1**, state the ranges of water depth over which duckweed and pondweed were found. [2]

Duckweed

Pondweed



Image 6.2 shows a photograph of duckweed and pondweed growing together and a diagram of their growth patterns.

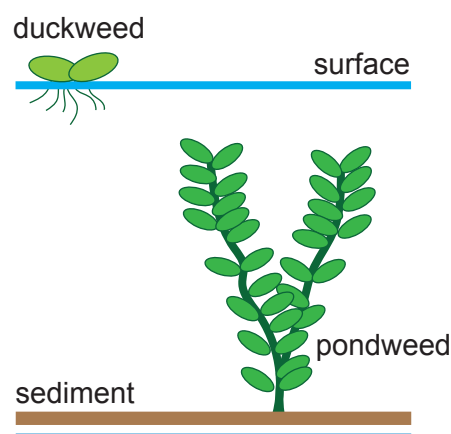
Image 6.2

duckweed growing
as a mat on the
surface of the water

pondweed growing
throughout the
water column



Growth patterns



- (b) (i) With reference to **Image 6.2** suggest why duckweed outcompetes pondweed in the shallower part of the range. [1]

.....

.....

- (ii) Name **two** nutrients for which the plants will compete **and** explain what each nutrient is used for in the plants. [3]

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- (iii) Explain why the concentration of nutrients is likely to be greater at the bottom of the pools and suggest a hypothesis to explain why pondweed outcompetes duckweed in the deeper water. [3]

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- (c) **Image 6.3** shows two snails that were found during the survey. Sampling revealed that the banded form was more common on dry land but the unbanded form was more common on the marsh.

Image 6.3



Cepaea nemoralis
banded form



Cepaea nemoralis
unbanded form

- (i) State how biologists would determine that the snails shown in **Image 6.3** belong to the same species. [1]

.....

.....

- (ii) Give the term used for the type of variation shown in **Image 6.3**. [1]

.....

- (iii) Suggest the advantage of bands to snails which live amongst twigs and leaves on the floor of the birch wood. [2]

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